

**Министр науки и высшего образования Российской
Федерации**

**Федеральное государственное автономное
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«Национальный исследовательский университет ИТМО»

**Факультет информационных технологий и
программирования**

Лабораторная работа № 4

«Создания процедуры наполнения Хранилища данных.»

Выполнил студент группы № М3315

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Цель работы: создать хранимую процедуру обеспечивающую автоматизированный перенос данных из Филиалов в Хранилище. Процедура должна обеспечить добавление вновь поступившей информации из Филиала Запад, и Филиала Восток в центральное хранилище данных. Процедура должна исключить дублирование информации.

Разработчик должен продумать механизм обеспечивающий минимализацию нагрузки на продуктивные сервера филиалов.

Скрипт:

```
CREATE OR REPLACE PROCEDURE public.load_storage()
```

```
LANGUAGE plpgsql
```

```
AS $load_storage$
```

```
BEGIN
```

```
    INSERT INTO public.DimCustomer (customername, entrydate, branchcustomerid, branchid)
```

```
        SELECT
```

```
            customername,
```

```
            NOW()::date AS entrydate,
```

```
            customerid AS branchcustomerid,
```

```
            (SELECT branchid
```

```
              FROM DimBranch
```

```
              WHERE branchname = 'West Branch')
```

```
        FROM west_schema.customer
```

```
        WHERE NOT EXISTS (
```

```
            SELECT 1 FROM DimCustomer dc
```

```
            WHERE dc.customername = west_schema.customer.customername
```

```
            AND dc.branchcustomerid = west_schema.customer.customerid
```

```
            AND dc.branchid = (SELECT branchid FROM DimBranch WHERE branchname = 'West Branch'))
```

```
        );
```

```
    INSERT INTO public.DimCategory (categoryid, categoryname, entrydate, branchcategoryid, branchid)
```

```
        SELECT uuid_generate_v4(), wc.categoryname, NOW()::date, wc.categoryid, db.branchid
```

```
        FROM west_schema.category wc
```

```
        JOIN DimBranch db ON db.branchname = 'West Branch'
```

WHERE NOT EXISTS (

SELECT 1 FROM DimCategory dc

WHERE dc.categoryname = wc.categoryname

AND dc.branchcategoryid = wc.categoryid

AND dc.branchid = db.branchid

);

INSERT INTO public.dimproduct (productId, productName, entryDate, branchProductId, branchId)

SELECT

uuid_generate_v4(),

productname,

CURRENT_DATE,

productid,

(SELECT branchId FROM public.dimbranch WHERE branchName = 'West Branch')

FROM west_schema.product

WHERE NOT EXISTS (

SELECT 1 FROM public.dimproduct dp

WHERE dp.productName = west_schema.product.productname

AND dp.branchProductId = west_schema.product.productid

AND dp.branchId = (SELECT branchId FROM public.dimbranch WHERE branchName = 'West Branch')

);

INSERT INTO public.dimtransaction (transactionId, transactionPrice, entryDate, branchTransactionId, branchId)

SELECT

uuid_generate_v4(),

totalamount,

CURRENT_DATE,

transactionid,

(SELECT branchId FROM public.dimbranch WHERE branchName = 'West Branch')

FROM west_schema.transaction

```

WHERE NOT EXISTS (

SELECT 1 FROM public.dimtransaction dt

WHERE dt.transactionPrice = west_schema.transaction.totalamount

AND dt.branchTransactionId = west_schema.transaction.transactionid

AND dt.branchId = (SELECT branchId FROM public.dimbranch WHERE branchName =

'West Branch')

);

```

END;

\$load_storage\$;

Query

Query History

1CALL public.load_storage();

2select * from dimcustomer

Data Output

Messages

Notifications

	customerid [PK] uuid	customername character varying (100)	entrydate date	branchcustomerid integer	branchid uuid
87	a0b7174d-9c4b-491e-879b-450794bf9f6c	Homer	2024-10-19	40	694b0f57-bb8b-4175-a44e-d69654ad1d33
88	19c7c79d-b203-4174-b784-1e795ad0fef8	Socrates	2024-10-19	41	694b0f57-bb8b-4175-a44e-d69654ad1d33
89	0aae8026-83f4-43bf-951d-e8fe31df1ac9	Plato	2024-10-19	42	694b0f57-bb8b-4175-a44e-d69654ad1d33
90	da355a0b-176b-4bc7-8262-ba88011ffd6b	Aristotle	2024-10-19	43	694b0f57-bb8b-4175-a44e-d69654ad1d33
91	4f2c5e1c-2484-46c4-8103-31a821f9b669	Alexander the Great	2024-10-19	44	694b0f57-bb8b-4175-a44e-d69654ad1d33
92	abe8dff4-b1e5-4b7f-8f6e-1f45509e844f	Genghis Khan	2024-10-19	45	694b0f57-bb8b-4175-a44e-d69654ad1d33
93	e339c321-1647-4079-b69f-42c614af6694	Napoleon Bonaparte	2024-10-19	46	694b0f57-bb8b-4175-a44e-d69654ad1d33
94	1bcc57aa-14ab-4208-ac21-abcbae4a8c...	Joan of Arc	2024-10-19	47	694b0f57-bb8b-4175-a44e-d69654ad1d33
95	125e542b-68d1-432d-8ac5-43283efb8e...	Cleopatra VII	2024-10-19	48	694b0f57-bb8b-4175-a44e-d69654ad1d33
96	f1476954-43d4-430d-a1cd-03bc68a79c...	Leon Trotsky	2024-10-19	49	694b0f57-bb8b-4175-a44e-d69654ad1d33
97	bfa34126-d2d5-4fd5-b79d-d8337da628...	Vladimir Lenin	2024-10-19	50	694b0f57-bb8b-4175-a44e-d69654ad1d33
98	8212b776-60ef-4c08-8548-8422e04a5fb1	Winston Churchill	2024-10-19	51	694b0f57-bb8b-4175-a44e-d69654ad1d33
99	a17b270d-91f0-4de2-81ec-de1cc7a1c1...	Snoop d	2024-11-11	52	694b0f57-bb8b-4175-a44e-d69654ad1d33
100	2ccffbd7-7ca6-4ee6-9d80-2e0278adeb34	2pac	2024-11-11	53	694b0f57-bb8b-4175-a44e-d69654ad1d33

Аналогично добавил из west branch

QueryQuery History

1CALL public.load_storage();

2select * from dimcustomer

Data OutputMessagesNotifications

	customerid [PK] uuid	customername character varying (100)	entrydate date	branchcustomerid integer	branchid uuid
94	1bcc57aa-14ab-4208-ac21-abcbac4a8c...	Joan of Arc	2024-10-19	47	694b0f57-bb8b-4175-a44e-d69654ad1d33
95	125e542b-68d1-432d-8ac5-43283efb8e...	Cleopatra VII	2024-10-19	48	694b0f57-bb8b-4175-a44e-d69654ad1d33
96	f1476954-43d4-430d-a1cd-03bc68a79c...	Leon Trotsky	2024-10-19	49	694b0f57-bb8b-4175-a44e-d69654ad1d33
97	bfa34126-d2d5-4fd5-b79d-d8337da628...	Vladimir Lenin	2024-10-19	50	694b0f57-bb8b-4175-a44e-d69654ad1d33
98	8212b776-60ef-4c08-8548-8422e04a5f...	Winston Churchill	2024-10-19	51	694b0f57-bb8b-4175-a44e-d69654ad1d33
99	a17b270d-91f0-4de2-81ec-de1cc7a1c1...	Snoop d	2024-11-11	52	694b0f57-bb8b-4175-a44e-d69654ad1d33
100	2ccffbd7-7ca6-4ee6-9d80-2e0278adeb...	2pac	2024-11-11	53	694b0f57-bb8b-4175-a44e-d69654ad1d33
101	602541f1-7840-44f2-8223-ade7417465...	Snoop d from west	2024-11-12	48	926c407e-1fa7-4778-852c-50cc671d9d9d
102	0721c307-7501-41f8-80f8-fbbfe57538b9	2pac from west	2024-11-12	49	926c407e-1fa7-4778-852c-50cc671d9d9d

QueryQuery History

1CALL public.load_storage();

2select * from dimcategory

Data OutputMessagesNotifications

	categoryid [PK] uuid	categoryname character varying (100)	entrydate date	branchcategoryid integer	branchid uuid
51	96350b09-d37a-469b-88d5-1ad95de485f9	Health & Wellness	2024-10-19	21	694b0f57-bb8b-4175-a44e-d69654ad1d33
52	8c192e22-8363-4ccc-9193-e32feaac0639	Home Improvement	2024-10-19	22	694b0f57-bb8b-4175-a44e-d69654ad1d33
53	a40eda2b-f150-4699-8310-2e9c05037920	Baby Products	2024-10-19	23	694b0f57-bb8b-4175-a44e-d69654ad1d33
54	b7420123-f856-44bb-b665-4ef4b9a09282	Stationery	2024-10-19	24	694b0f57-bb8b-4175-a44e-d69654ad1d33
55	f15b2473-9983-4c2e-bbe5-96be94de52d5	Jewelry	2024-10-19	25	694b0f57-bb8b-4175-a44e-d69654ad1d33
56	c478cd8a-235e-4bf4-999d-1cd74a3c79a3	Gadgets	2024-10-19	26	694b0f57-bb8b-4175-a44e-d69654ad1d33
57	c4aaf6a9-8c09-4b47-b966-a42f4fe29c95	Gaming	2024-10-19	27	694b0f57-bb8b-4175-a44e-d69654ad1d33
58	07e9e83f-ba22-4768-8a9e-f0abbd45f480	Smart Devices	2024-10-19	28	694b0f57-bb8b-4175-a44e-d69654ad1d33
59	def139f3-f3b9-48d4-9ffa-7a9fc085c9f3	Cleaning Equipment	2024-10-19	29	694b0f57-bb8b-4175-a44e-d69654ad1d33
60	abbc3b2d-0fad-41e0-b9d1-fd4bd692633a	Groceries	2024-10-19	30	694b0f57-bb8b-4175-a44e-d69654ad1d33
61	b915bc49-6ac9-4940-9091-6853b323f0b8	Computers	2024-11-12	31	694b0f57-bb8b-4175-a44e-d69654ad1d33

Query

Query History

1

CALL public.load_storage();

s

2

select * from dimproduct

Data Output

Messages

Notifications

	productid [PK] uuid	productname character varying (100)	entrydate date	branchproductid integer	branchid uuid
91	33bc4996-d4c2-4f39-9d51-cd344eb127c5	VR Headset	2024-10-19	41	694b0f57-bb8b-4175-a44e-d69654ad1d33
92	c1c32457-13ea-424f-9158-0ad18c42a890	3D Printer	2024-10-19	42	694b0f57-bb8b-4175-a44e-d69654ad1d33
93	ec5575ed-dfb8-4ccd-9456-c0510b1e9be8	Portable Projector	2024-10-19	43	694b0f57-bb8b-4175-a44e-d69654ad1d33
94	0cab1ac3-c705-4411-b674-97249a80be41	Smart Glasses	2024-10-19	44	694b0f57-bb8b-4175-a44e-d69654ad1d33
95	68c2e656-123f-4bf7-aa7f-b7d4e9f015ee	Electric Skateboard	2024-10-19	45	694b0f57-bb8b-4175-a44e-d69654ad1d33
96	45b8f5e8-540a-4831-92ad-73382ecbe09a	Hoverboard	2024-10-19	46	694b0f57-bb8b-4175-a44e-d69654ad1d33
97	5105ddaa-b1c4-4e18-8e1e-d897f2ca1e14	Electric Bike	2024-10-19	47	694b0f57-bb8b-4175-a44e-d69654ad1d33
98	d2ed24c6-1747-4e57-85c7-a343bbc30f96	Electric Scooter	2024-10-19	48	694b0f57-bb8b-4175-a44e-d69654ad1d33
99	82742a08-ecfe-433c-9e41-ec95730396d1	Smart Thermostat	2024-10-19	49	694b0f57-bb8b-4175-a44e-d69654ad1d33
100	9c0337af-c3d7-4c94-9352-baad6b6925c4	Air Purifier	2024-10-19	50	694b0f57-bb8b-4175-a44e-d69654ad1d33
101	81833638-0762-40db-a412-7e337442cf47	Lays	2024-11-12	51	694b0f57-bb8b-4175-a44e-d69654ad1d33

Query

Query History

1

CALL public.load_storage();

2

select * from dimtransaction

Data Output

Messages

Notifications

	transactionid [PK] uuid	transactionprice numeric (10,2)	entrydate date	branchtransactionid integer	branchid uuid
89	8f86a014-a581-4bb8-913d-1f1dbf7d48c0	400.00	2024-10-19	42	694b0f57-bb8b-4175-a44e-d69654ad1d33
90	73270d39-211f-4b69-9849-f70a662c06d4	250.00	2024-10-19	43	694b0f57-bb8b-4175-a44e-d69654ad1d33
91	f842e9f6-ca83-4120-b3ef-ffb98339473c	150.00	2024-10-19	44	694b0f57-bb8b-4175-a44e-d69654ad1d33
92	7b4774d8-dc8d-4d2d-b27a-8649a0c86f1c	700.00	2024-10-19	45	694b0f57-bb8b-4175-a44e-d69654ad1d33
93	9f4ac6a3-1771-481e-bbcb-16b7eccf2624	300.00	2024-10-19	46	694b0f57-bb8b-4175-a44e-d69654ad1d33
94	36462ac0-a5ea-4da7-aafo-9a5f3f0b95ef	500.00	2024-10-19	47	694b0f57-bb8b-4175-a44e-d69654ad1d33
95	0bd1413b-4402-487f-b321-c82affed474f	750.00	2024-10-19	48	694b0f57-bb8b-4175-a44e-d69654ad1d33
96	9e29d1fa-756e-41bb-81b4-53e1a8741aaa	400.00	2024-10-19	49	694b0f57-bb8b-4175-a44e-d69654ad1d33
97	0f07b513-80bf-477c-9279-38e2341768f0	600.00	2024-10-19	50	694b0f57-bb8b-4175-a44e-d69654ad1d33
98	d83290e5-9e38-4484-8985-bcf8616a098e	500.00	2024-11-12	51	694b0f57-bb8b-4175-a44e-d69654ad1d33
99	95c632b5-c130-4815-b672-318780d36fe3	300.00	2024-11-12	52	694b0f57-bb8b-4175-a44e-d69654ad1d33